

E3 — formal conformance, STATIC part (dynamic checks reserved for E4)

Obligation. Compare displayed theorems/quantitative corollaries with Lean types; check edge-partition semantics, chordality, gain conventions, coercions, strict inequalities, root predicates, threshold dependence, discharge of premises.

Inputs. LEAN_SOURCE_SNAPSHOT_v1.0.zip sha256 0a13c01c...f8a7dd (extracted to C:/piv_r2/src, 504 .lean files matching SOURCES.json f1769273..., see E0/E0_sources_vs_zip.json); annex ZIP 2847a422...837d (internal SOURCE_MANIFEST.sha256: 39/39 OK after CRLF normalisation); EN manuscript md 71417794....

Independent method. Auditor-written extractor `extract_headers.py` (literal text of each header up to `:=`), manual reading of every definition occurring in the headline types (outputs: `headers_part1.txt`, `headers_part2.txt`, `defs_part1.txt`, `defs_part2.txt`), textual sweep for escape hatches (`textual_sweep_counts.txt`, `textual_sweep_hits.txt`), existence check of every backticked identifier in EN/ES (`name_check.py` → `name_check.json`). The author's audit scripts were **not** used for these results.

Findings (static)

Item	Result								
Partition semantics	<code>CliquePartition</code> : pieces are cliques, $\text{card} \geq 2$, distinct pieces have disjoint <code>pairs.biUnion pairs = edgeFinset</code> (exact edge partition). <code>OrderAtMost r</code> : all pieces $\text{card} \leq r$. <code>size</code> = number of pieces. MATCH with §1/§2.								
Chordality	<code>SimpleGraph.IsChordal</code> : every cycle of length ≥ 4 has two support vertices adjacent by a non-cycle edge (a chord). <code>PaperIV.FarRounding.IsChordal</code> is an abbrev of it. MATCH.								
Target	<code>targetSize n = n*(n+1)/6</code> ($\mathbb{N}$ division = floor). MATCH <code>M(n)</code> .								
Thm A / B / max	headers quantify $\exists b, \forall n$ (incl. 0) $\forall G$ resp. $\exists N \forall n \geq N$; lower witness optimal against all <code>CliquePartition</code> (no order restriction). MATCH.								
Thm 3.1	<code>UniformRoundingTarget ε</code> : $\exists N, \forall n \geq N, \forall G, \forall x$: <code>FracPacking G (Q)</code> , $\exists P$: <code>Packing G, x.value - P.gain $\leq \varepsilon n^2$</code> ; items = cliques of card $3/4$, gain $C(k,2)-1$. No chordality. MATCH, incl. quantifier order.								
Thm 5.0	hypotheses $n \geq 4 \cdot 10^{12}$, chordal, certified optimum, normalized distance $< \varepsilon_0$ to all labelled complete-split supports (cores = all subsets), near value; conclusion clique <code>R,</code>	3	R	-n	$\leq n/50, \text{ order} \leq 4,$	Q	$\leq B_n($	R	) - m/ - A/2' MATC
Thm 6.1	$\gamma = \text{eta}/4 = 1/(4 \cdot 10^{16})$ with <code>gamma_pos</code> ; $\delta \in \mathbb{Q}, 0 \leq \delta \leq \gamma n^2$; hypothesis quantifies over order- ≤ 4 partitions; conclusion: clique <code>R, 2 $\leq$</code>	R	$\leq n-$	R	$, B \leq M, (117/1825) m + (12687/20000) A \leq \delta, (M - B) + m/16 + A/2 \leq \delta,$ <code>editDist(G, S_R) $\leq 16 \delta$</code> . MATCH. Manuscript's "there exist $\gamma > 0$ " is realised by an explicit constant. δ rational vs. real: harmless (integrality allows replacing real δ by $\lfloor \delta \rfloor$) — OBSERVATION only.				

Cor 6.1a	$\exists R \text{ clique}, 2 \leq$	R	$\leq n-$	R	$\wedge \forall (Q : \text{CliquePartition } G) (\tau : \mathbb{Q}),$ Q.size $\leq M + \tau \rightarrow \# \text{noncanonical} \leq \tau$ + 48 δ . Root precedes every partition; no order restriction on Q; τ unrestricted (stronger than the manuscript's $\tau \geq$ 0) . IsCanonicalAt: exactly 1 outside vertex and 1 or 2 root vertices. MATCH (Lean stronger).
Cor 6.2	CliqueCoverEq G M $\leftrightarrow$ CliqueCover4Eq G M and $\text{CliqueCover4Eq } G \ M \leftrightarrow \exists R, G =$ splitGraph R (univ\R) $\wedge$ OptimalCore n	R	; OptimalCore n k $\leftrightarrow$ k= $\lfloor (n+1)/3 \rfloor \vee$ ($n \equiv 1 \bmod 3 \wedge$ k= $\lfloor (n+1)/3 \rfloor + 1$) = (6.4). MATCH.		
Thm C	RoutedDefectAt G s: $\forall U \ R, R \subseteq U,$ R clique, $U \cap R \neq \emptyset \rightarrow \exists v \in U \cap R, \exists \text{ clique}$ $C \subseteq N_U(v),$	$N_U(v)$	$\leq$	C	+ s. Empty root allowed. Target defectTarget s n = targetSize (n+s) - C(s+1, 2) (truncated $\mathbb{N}$ subtraction, irrelevant for $n \geq s+1$ by targetSize_le_defectTarget) Maximum witness optimal vs. unrestricted partitions. MATCH.
Prop 6.3	Lean gives the fractional upper bound $x.\text{value} \leq 2C(k, 2)$ for all fractional packings and a clique partition of total (unrestricted) gain $2C(k, 2)$; the equality $W = W^*$ as a statement about mixed packings is obtained by combination with SplitCompleteSharpValue, not literally one declaration. OBSERVATION.				
Explicit tower	E18.Numeric.hIter literal = manuscript h; E19.tower2 0 = 1, tower2 (m+1) = $2^{\text{tower2 } m} = T$; sharpThreshold = tower2 (hIter+7); b $\leq$ tower2 (hIter+8). MATCH.				
Annex (8.1a)	chordal_gap_linear_cliqueFree: chordal, CliqueFree (d+2), any fractional triangle packing: value $\leq v_3 +$ (10 + d/2)	V	. Triangular only; not in import PaperIV. MATCH.		
Escape hatches (text)	0 code occurrences of sorry/admit/ axiom declarations/native_decide/ implemented_by/unsafe/opaque/ extern/ofReduceBool/ skipKernelTC — all hits are in comments/docstrings. Metaprograms (elab, #eval ... CoreM) exist only in audit modules (E17/*Audit, ConeAudit, PublicationIncrementAudit, ExplicitThresholdAudit, ConstructorBudgetAudit, E19/ AxiomCheck, A4S1/IndepAllAudit); none uses addDecl. Dynamic #print axioms reserved for E4.				
Identifier existence	All backticked Lean identifiers in EN/ES resolve to modules/declarations in the freeze, except annex names (present in annex) and non-Lean tokens (hashes, file names, namespaces, Mathlib's Behrend.roth_lower_bound, [5]'s Erdos81/ ExternalInputs.Inputs).				

Negative control. The extractor was run on a deliberately non-existent name (OrderAtMost as a top-level decl: 0 hits, correctly, since it is CliquePartition.OrderAtMost) — demonstrates the extractor reports absence rather than

fabricating a match.

Static result. No manuscript/Lean mismatch found for the headline statements. E3 overall remains **pending E4** (types/axioms must be confirmed by the kernel in the isolated build; #check of the 195 exports and cone inspection are dynamic).

Limitations. Static text matching cannot detect notation/instance tricks that change meaning at elaboration; this is why E4 re-prints the elaborated types.

Final status (2026-09-28)

E4 was not run (STOP at the pre-build checkpoint). The static comparison found **no mismatch**, but the dynamic confirmation of elaborated types, axioms, cones and the 195 exports is missing. **E3 = INCONCLUSIVE.**

Rendered from the Markdown semantic source with PyMuPDF; not compiled from the TeX file (no TeX engine installed).

